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```
[6] # =====
# ITA0613 MACHINE LEARNING ASSIGNMENT
# Indian Historical Crop Yield & Weather Dataset
# ONE-CELL COMPLETE IMPLEMENTATION
# =====

!pip -q install scipy

import os, glob, zipfile, requests, warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from scipy.spatial.distance import mahalanobis
from scipy.linalg import pinv

print("="*70)
print("DOWNLOADING INDIAN CROP YIELD + WEATHER DATASET")
print("="*70)

# -----
# 1. DOWNLOAD DATASET DIRECTLY FROM KAGGLE
# -----

url = "https://www.kaggle.com/api/v1/datasets/download/zoya77/indian-historical-crop-yield-and-weather-data"

zip_path = "/content/indian_crop_data.zip"

r = requests.get(url)
open(zip_path, "wb").write(r.content)

extract_path = "/content/indian_crop_dataset"

os.makedirs(extract_path, exist_ok=True)

with zipfile.ZipFile(zip_path, "r") as z:
```

RAM

Disk

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{} Variables

Terminal

⬇

Executing (23s)

Python 3

🔍 Variables
📄 Terminal
⚙️
📊 Executing (5%)
🐍 Python 3

... RAM []
Disk []

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```
# -----
# 5. AUTOMATIC COLUMN DETECTION
# -----

def find_col(columns, keywords):

    for c in columns:
        n = norm(c)

        for k in keywords:
            if k in n:
                return c

    return None

year_col = find_col(df.columns,
                    ["year", "yr"])

yield_col = find_col(df.columns,
                    ["yield", "yields", "yieldsper"])

rain_col = find_col(df.columns,
                    ["rainfall", "rain"])

temp_col = find_col(df.columns,
                    ["temperature", "temp"])

humidity_col = find_col(df.columns,
                       ["humidity", "humid"])

wind_col = find_col(df.columns,
                    ["wind"])

solar_col = find_col(df.columns,
                     ["solar", "radiation"])

state_col = find_col(df.columns,
                     ["state"])

district_col = find_col(df.columns,
                        ["district"])

crop_col = find_col(df.columns,
                    ["crop"])

print("\nDETECTED COLUMNS")
print("Year      :", year_col)
print("Yield      :", yield_col)
print("Rainfall    :", rain_col)
print("Temperature:", temp_col)
print("Humidity    :", humidity_col)
print("Wind       :", wind_col)
print("Solar       :", solar_col)
print("State       :", state_col)
print("District    :", district_col)
```

Variables

Terminal

Executing (1m 1s) Python 3

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Variables
Terminal
Python 3
Executing (1m 9s)

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```
if rain_col:
    df["Rainfall_Squared"] = df[rain_col] ** 2
```

```
if rsin_col and temp_col:
    #["Rainfall Temperature Ratio"] = (
```

```
if temp_col and humidity_col:
    df["Temperature Humidity Index"] = (
```

```
if rain_col:
    df["Rainfall Anomaly"] = (
```

```
print("\n===== DATASET SUMMARY =====")
```

```
plt.figure(figsize=(8,5))
plt.hist(df[yield_col].dropna(), bins=30)
plt.xlabel('Yield (kg/ha)')
plt.ylabel('Frequency')
```

```
if rain_col:
    plt.figure(figsize=(8,5))
```

Variables
Terminal
Executing (1m 31s)
Python 3

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```
if train_mask.sum() > 0 and val_mask.sum() > 0 and test_mask.sum() > 0:
    X_train = X[train_mask.values]
    y_train = y[train_mask.values]
    X_val = X[val_mask.values]
    y_val = y[val_mask.values]
    X_test = X[test_mask.values]
    y_test = y[test_mask.values]
else:
    idx = np.arange(n)
    np.random.seed(42)
    np.random.shuffle(idx)
    a = int(.6*n)
    b = int(.8*n)
    X_train, y_train = X[idx[a:]] , y[idx[a:]]
    X_val, y_val = X[idx[a:b]] , y[idx[a:b]]
    X_test, y_test = X[idx[b:]] , y[idx[b:]]
else:
    idx = np.arange(n)
    np.random.seed(42)
    np.random.shuffle(idx)
    a = int(.6*n)
    b = int(.8*n)
    X_train, y_train = X[idx[a:]] , y[idx[a:]]
    X_val, y_val = X[idx[a:b]] , y[idx[a:b]]
    X_test, y_test = X[idx[b:]] , y[idx[b:]]
print("\nTRAIN:", X_train.shape)
print("\nVALIDATION:", X_val.shape)
print("\nTEST:", X_test.shape)
# -----
# 17. KNN FROM FIRST PRINCIPLES
# -----
def knn_predict(Xtr, ytr, Xte, k=6, metric="euclidean"):
    predictions = []
    if metric == "mahalanobis":
        cov = np.cov(Xtr, rowvar=False)
        VI = pinv(cov + np.eye(cov.shape[0])*1e-6)
        for x in Xte:
            if metric == "euclidean":
```

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Variables Terminal

Executing (tm 46s) Python 3

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```
def knn_predict(Xtr, ytr, Xte, k=5, metric="euclidean"):
    predictions = []
```

```

cov = np.cov(XTr, rowvar=False)
VI = pinv(cov + np.eye(cov.shape[0]))*1e-6)

```

```
if metric == "euclidean":
```

$$(x \cdot x) \otimes 1 \otimes 1 \otimes (x \cdot x)$$

```
nearest = np.argsort(distances)[:k]
nearest_distances = nearest
```

```
return np.array(predictions)
```

validation errors = 0

Variables Terminal Executing (2m 10s) Python 3

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23. CANDIDATE ELIMINATION

```
# -----  
risk = np.where(  
    y < np.quantile(y,0.33),  
    "Low",  
    np.where(  
        y < np.quantile(y,0.67),  
        "Medium",  
        "High"  
    )  
)  
  
candidate_data = df[features].copy()  
candidate_data["Risk"] = risk  
  
print("\n===== YIELD RISK CLASSES =====")  
print(  
    candidate_data["Risk"]  
    .value_counts()  
)  
  
# Simple version-space style boundary discovery  
positive = candidate_data[  
    candidate_data["Risk"] == "High"  
]  
  
negative = candidate_data[  
    candidate_data["Risk"] != "High"  
]  
  
S = {}  
  
for c in features:  
    if len(positive)>0:  
        S[c] = {  
            positive[c].min(),  
            positive[c].max()  
        }  
  
print("\nvCandidate-Elimination style specific boundary.")  
print(S)  
  
# -----  
24. SCALABILITY EXPERIMENT  
# -----  
  
sizes = [1000,5000,10000]  
times=[]  
  
import time
```

RAM
Disk

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Variables Terminal

Executing (2m 36s) Python 3

```
def random_hash(x, bits=8, seed=42):
    rng = np.random.default_rng(seed)

    planes = rng.normal(
        size=(bits, x.shape[1])
    )

    signatures = (X @ planes.T >= 0).astype(int)
```


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=====

DOWNLOADING INDIAN CROP YIELD + WEATHER DATASET

=====

Dataset downloaded successfully.

CSV files found:

- Custom_Crops_yield_Historical_Dataset.csv

Custom_Crops_yield_Historical_Dataset.csv (50765, 20)

['Dist Code', 'Year', 'State Code', 'State Name', 'Dist Name', 'Crop', 'Area_ha', 'Yield_kg_per_ha', 'N_req_kg_per_ha', 'P_req_kg_per_ha', 'K_req_kg_per_ha', 'Total_N_kg', 'Total_P_kg', 'Total_K_kg', 'Temperature_C', 'Humidity_N', 'pH', 'Rainfall_mm', 'Wind_Speed_m_s', 'Solar_Radiation_MJ_m2_day']

FINAL DATASET SHAPE: (50765, 20)

COLUMNS:

['Dist Code', 'Year', 'State Code', 'State Name', 'Dist Name', 'Crop', 'Area_ha', 'Yield_kg_per_ha', 'N_req_kg_per_ha', 'P_req_kg_per_ha', 'K_req_kg_per_ha', 'Total_N_kg', 'Total_P_kg', 'Total_K_kg', 'Temperature_C', 'Humidity_N', 'pH', 'Rainfall_mm', 'Wind_Speed_m_s', 'Solar_Radiation_MJ_m2_day']

DETECTED COLUMN

Year : Year

Yield : Yield_kg_per_ha

Rainfall : Rainfall_mm

Temperature : Temperature_C

Humidity : Humidity_N

Wind : Wind_Speed_m_s

Solar : Solar_Radiation_MJ_m2_day

State : State Code

District : None

Crop : Crop

Missing values after cleaning:

Dist Code 0

Year 0

State Code 0

State Name 0

Dist Name 0

Crop 0

Area_ha 0

Yield_kg_per_ha 0

N_req_kg_per_ha 0

P_req_kg_per_ha 0

K_req_kg_per_ha 0

dtype: int64

Engineered features created.

----- DATASET SUMMARY -----

Dist Code count unique Too freq \

Year 50765.0 NaN NaN NaN

State Code 50765.0 NaN NaN NaN

State Name 50765 20 Uttar Pradesh 7488

Dist Name 50765 311 Khargone / West Nimar 298

Crop 50765 4 Rice 14978

Area_ha 50765.0 NaN NaN NaN

Yield_kg_per_ha 50765.0 NaN NaN NaN

N_req_kg_per_ha 50765.0 NaN NaN NaN

P_req_kg_per_ha 50765.0 NaN NaN NaN

K_req_kg_per_ha 50765.0 NaN NaN NaN

Temperature_C 50765.0 NaN NaN NaN

Humidity_N 50765.0 NaN NaN NaN

pH 50765.0 NaN NaN NaN

Rainfall_mm 50765.0 NaN NaN NaN

Wind_Speed_m_s 50765.0 NaN NaN NaN

Solar_Radiation_MJ_m2_day 50765.0 NaN NaN NaN

Variables

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===== DATASET SUMMARY =====

	count	unique	top	freq
Dist Code	50765.0	NaN	NaN	NaN
Year	50765.0	NaN	NaN	NaN
State Code	50765.0	NaN	NaN	NaN
State Name	50765	20	Uttar Pradesh	7455
Dist Name	50765	313	Kharagone / West Bihar	288
Crop	50765	4	rice	14978
Area_ha	50765.0	NaN	NaN	NaN
Yield_kg_per_ha	50765.0	NaN	NaN	NaN
N_req_kg_per_ha	50765.0	NaN	NaN	NaN
P_req_kg_per_ha	50765.0	NaN	NaN	NaN
K_req_kg_per_ha	50765.0	NaN	NaN	NaN
Total_L_kg	50765.0	NaN	NaN	NaN
Total_P_kg	50765.0	NaN	NaN	NaN
Total_K_kg	50765.0	NaN	NaN	NaN
Temperature_C	50765.0	NaN	NaN	NaN
Humidity_%	50765.0	NaN	NaN	NaN
pH	50765.0	NaN	NaN	NaN
Rainfall_mm	50765.0	NaN	NaN	NaN
Wind_Speed_m_s	50765.0	NaN	NaN	NaN
Solar_Radiation_MJ_m2_day	50765.0	NaN	NaN	NaN

	mean	std	min
Dist Code	239.218753	209.862837	1.0
Year	1991.56277	14.336144	1966.0
State Code	9.169822	4.852893	1.0
State Name	NaN	NaN	NaN
Dist Name	NaN	NaN	NaN
Crop	NaN	NaN	NaN
Area_ha	63785.22683	112798.17884	10.0
Yield_kg_per_ha	1159.814711	1805.866807	0.28
N_req_kg_per_ha	20.567342	16.389891	0.00756
P_req_kg_per_ha	13.537313	12.836334	0.00336
K_req_kg_per_ha	22.676814	19.432144	0.00756
Total_L_kg	2263185.592686	532256.797286	9.18
Total_P_kg	1888930.30387	2544754.821754	4.08
Total_K_kg	1033638.064321	4648001.426882	0.18
Temperature_C	23.261972	2.772772	20.0
Humidity_%	69.454151	7.76709	68.0
pH	6.283986	0.247693	6.0
Rainfall_mm	848.89219	239.465323	680.0
Wind_Speed_m_s	2.158574	0.516744	1.5
Solar_Radiation_MJ_m2_day	18.626298	2.866976	16.0

	25%	50%	75%	max
Dist Code	72.0	164.0	239.0	917.0
Year	1979.0	1992.0	2004.0	2017.0
State Code	6.0	9.0	12.0	20.0
State Name	NaN	NaN	NaN	NaN
Dist Name	NaN	NaN	NaN	NaN
Crop	NaN	NaN	NaN	NaN
Area_ha	2340.0	36980.0	68080.0	1154259.0
Yield_kg_per_ha	500.0	922.88	1533.6	58363.64
N_req_kg_per_ha	10.44702	20.29131	39.57471	1050.54552
P_req_kg_per_ha	5.36312	10.8387	18.10864	583.6364
K_req_kg_per_ha	9.72092	17.0	29.66282	1050.54552

Variables Terminal

Executing (3m 32s) Python 3

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Dataset size: (56765, 25)

	25%	50%	75%	max
Dist Code	72.0	144.0	238.0	917.0
Year	1979.0	1992.0	2004.0	2017.0
State Code	6.0	9.0	12.0	20.0
State Name	NaN	NaN	NaN	NaN
Dist Name	NaN	NaN	NaN	NaN
Crop	NaN	NaN	NaN	NaN
Area_ha	2340.0	36980.0	68980.0	1154230.0
Yield_kg_per_ha	580.0	922.08	1533.6	58363.64
Area_kg_per_ha	10.44702	28.29131	39.57471	1050.54552
P_rn_kg_per_ha	5.3612	16.8967	18.18864	585.6364
K_rn_kg_per_ha	9.72092	17.0	29.46282	1059.54552
Total_kg	15928.4064	250000.0	1692500.4	80375238.06
Total_P_kg	17040.936	140000.1	808144.8836	38580114.2688
Total_K_kg	30239.6976	241998.68	1178627.0246	70730209.4928
Temperature_C	20.0	22.0	25.0	28.0
Humidity_%	60.0	70.0	80.0	88.0
pH	6.0	6.5	6.5	6.5
RainFall_mm	600.0	800.0	1200.0	1200.0
Wind_Speed_m_s	1.5	2.0	2.5	3.0
Solar_Radiation_MJ_m2_day	16.0	18.0	20.0	22.0

Distribution of Crop Yield

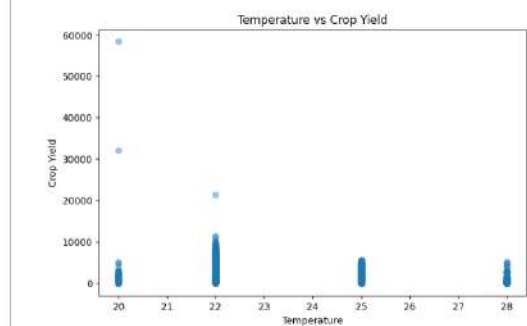
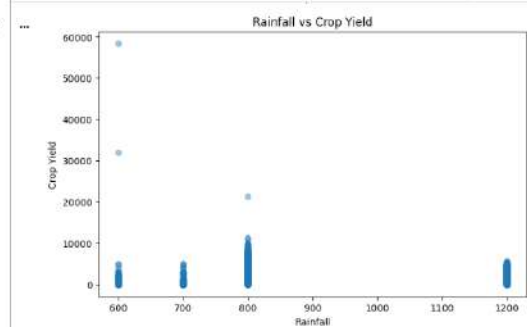
Rainfall vs Crop Yield

Variables

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Average Crop Yield Over Years

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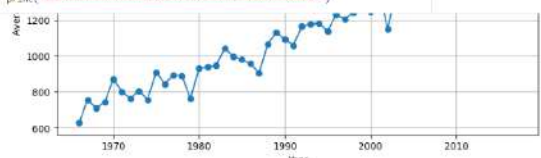
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colab.research.google.com/drive/1DlipoIFX9OWIKgRtK4pazmGyixesOpci#scrollTo=dKDRH-AndvpY8

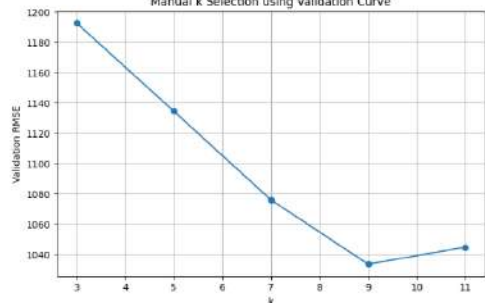
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print("-----")
print("PIPELINE COMPLETED SUCCESSFULLY")
print("-----")



TRAIN: (31217, 10)
VALIDATION: (9928, 10)
TEST: (9728, 10)
k = 3 Validation RMSE = 1102.4751
k = 5 Validation RMSE = 1134.3114
k = 7 Validation RMSE = 1075.8819
k = 9 Validation RMSE = 1033.491
k = 11 Validation RMSE = 1046.6799
BEST K = 9



----- (RM, BUILDING) -----
RMSE : 1114.1795719840895
MAE : 699.3641059827777
R2 : 0.279997669154486

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